

2015

Year 12 Extension 1 Mathematics

Trial Examination

Teacher Setting Paper: Mrs M Hill

Head of Department: Mrs M Hill

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board approved calculator may be used
- Write your answers to Section 1 on the multiple choice answer sheet provided
- Write your answers to Section 2 in the answer booklets provided. Start a new booklet for each question
- Write your student number only on the front of each booklet
- A table of standard integrals is provided at the back of this paper
- In Questions 11-14, show relevant mathematical reasoning and/or calculations

Total marks – 70

Section 1 Pages 2-3

10 marks

- Attempt all questions 1-10
- Allow 15 minutes for this section

Section 2 Pages 4-8

60 marks

- Attempt all questions 11-14
- Allow 1 hour and 45 minutes for this section

This examination paper does not necessarily reflect the content or format of the Higher School Certificate Examination in this subject

Section 1

10 marks

Attempt Questions 1 -10

Allow 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 -10

1. Which of the following is an expression for $\int x\sqrt{1-x^2} dx$?

Use the substitution $u = 1 - x^2$.

- (A) $-\frac{(1-x^2)^3}{3} + c$ (B) $\frac{(1-x^2)^3}{3} + c$
(C) $-\frac{(1-x^2)^{\frac{3}{2}}}{3} + c$ (D) $\frac{(1-x^2)^{\frac{3}{2}}}{3} + c$

2. Which of the following is an expression for $\int \sin^2 6x dx$

- (A) $\frac{x}{2} - \frac{1}{12} \sin 6x + c$ (B) $\frac{x}{2} + \frac{1}{12} \sin 6x + c$
(C) $\frac{x}{2} - \frac{1}{24} \sin 12x + c$ (D) $\frac{x}{2} + \frac{1}{24} \sin 12x + c$

3. A particle is moving along the x -axis. Its velocity v at position x is given by $v = \sqrt{8x - x^2}$.
What is the acceleration when $x=3$?

- (A) 1 (B) 2
(C) 3 (D) 4

4. A class consists of 15 students, of whom 5 are prefects. How many committees of 8 students can be formed if each committee contains exactly 2 prefects?

- (A) ${}^5C_2 \times {}^{10}C_6$ (B) ${}^5P_2 \times {}^{10}P_6$
(C) ${}^{15}C_8$ (D) $2! \times {}^{10}C_6$

5. A particle moves in a straight line and its position at any time t is given by
 $x = 3 \cos 2t + 4 \sin 2t$. The motion is simple harmonic. What is the greatest speed?

- (A) 6 (B) 10
(C) 12 (D) 20

6. If $f(x) = e^{x+2}$ what is the inverse function $f^{-1}(x)$?

(A) $f^{-1}(x) = e^{y-2}$

(B) $f^{-1}(x) = e^{y+2}$

(C) $f^{-1}(x) = \log_e x + 2$

(D) $f^{-1}(x) = \log_e x - 2$

7. What is the domain and range of $y = \cos^{-1}(\frac{3x}{2})$?

(A) Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$; Range: $0 \leq y \leq \pi$

(B) Domain: $-1 \leq x \leq 1$; Range: $0 \leq y \leq \pi$

(C) Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$; Range: $-\pi \leq y \leq \pi$

(D) Domain: $-1 \leq x \leq 1$; Range: $-\pi \leq y \leq \pi$

8. What is the exact value of the definite integral $\int_{\frac{2}{\sqrt{3}}}^{\frac{2\sqrt{3}}{3}} \frac{dx}{x^2 + 4}$?

(A) $\frac{\pi}{12}$

(B) $\frac{\pi}{6}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

9. What is the term independent of x in the expansion of $(x^2 - \frac{2}{x})^9$?

(A) ${}^9C_3(-2)^3$

(B) ${}^9C_6(-2)^6$

(C) ${}^9C_3(2)^3$

(D) ${}^9C_6(2)^6$

10. The function $f(x) = \sin x - \frac{2x}{3}$ has a real root close to $x = 1.5$.

Let $x = 1.5$ be a first approximation to the root.

What is the second approximation to the root using Newton's method?

(A) 1.495

(B) 1.496

(C) 1.503

(D) 1.504

Section 2

60 marks

Attempt Questions 11-14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet

- a) Solve for x : $\frac{4}{5-x} \geq 1$ 3
- b) Find the coordinates of the point which divides the interval AB with $A(1,4)$ and $B(5,2)$ externally in the ratio $1:3$. 2
- c) Let all the different arrangement of all the letters of DELETED be called a word.
- i) How many words are possible? 1
- ii) In how many words of these words will the D's be separated? 2
- d) Find the acute angle between the straight lines $y = 2x$ and $x + y - 3 = 0$. Give answer to the nearest minute. 3
- e) If $P(x) = x^3 - 2x^2 + ax + 4$ is divisible by $(x + 2)$, what is the value of a ? 2
- f) The polynomial equation $3x^3 - 2x^2 + 3x - 4 = 0$ has roots α, β, γ . Find the exact value of $\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha}$. 2

Question 12 (15 marks) Use a SEPARATE writing booklet

a) i) Find $\frac{d}{dx} \ln(\cos 2x)$ 1

ii) Hence evaluate exactly $\int_0^{\frac{\pi}{6}} \tan 2x \, dx$ 2

b) For a series $T_{n+1} - T_n = 7$ and $T_1 = 3$. Find the value of S_{100} , where $S_n = T_1 + T_2 + \dots + T_n$. 2

c) i) Write $\cos x - \sqrt{3} \sin x$ in the form $A \cos(x + \alpha)$ where $A > 0$, $0 < \alpha < \pi$. 2

ii) Hence or otherwise, solve $\cos x - \sqrt{3} \sin x = 1$ for all values of x . 2

d) Prove by mathematical induction that if n is a positive integer, then:

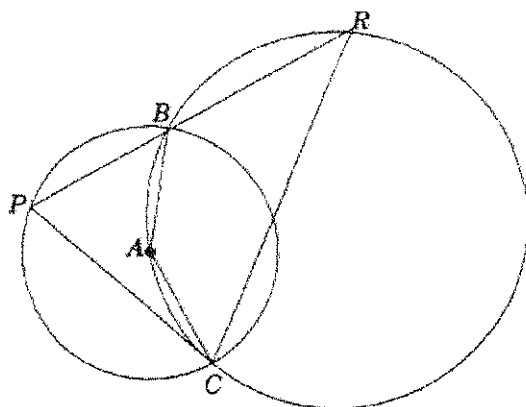
$$\frac{1}{1 \times 5} + \frac{1}{5 \times 9} + \frac{1}{9 \times 13} + \dots + \frac{1}{(4n-3)(4n+1)} = \frac{n}{4n+1}$$
 3

e) i) Sketch the graph of the function $y = \log_e(x-2)$ 1

ii) The region bounded by the curve $y = \log_e(x-2)$, the y -axis, $y=0$ and $y=h$ is rotated about the y -axis to create a bowl. Show that the volume of the bowl, V , is 2
 given by: $V = \pi \left(\frac{e^{2h}}{2} + 4e^h + 4h - \frac{9}{2} \right)$

Question 13 (15 marks) Use a SEPARATE writing booklet

a)



**Diagram is not
to scale**

A is the centre of the circle BCP . The point A lies on another circle BAC . The two circles intersect in B and C as shown in the diagram. PBR is a straight line. 3

Copy or trace this diagram into your writing booklet.

Prove, with reasons, that $RP = RC$.

b) From the top of a mountain 200 metres above ground an observer sights two landmarks A and B . Point A has a bearing of $300^\circ T$ at an angle of depression of 10° . Point B has a bearing of $040^\circ T$ at an angle of depression of 15° . Both points A and B are at ground level.

- i) Draw and label a diagram showing all the information 1
- ii) Calculate the distance from A to B (give answer to the nearest metre) 3

c) Let T be the temperature inside a room at time t hours and let A be the constant outside air temperature. Newton's Law of Cooling states that the rate of change of the temperature T is proportional to $(T - A)$.

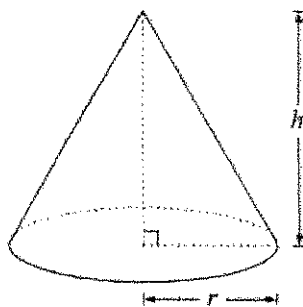
- i) Show that $T = A + Ce^{kt}$ where C and k are constants satisfies Newton's Law of Cooling $\frac{dT}{dt} = k(T - A)$. 1
- ii) The outside temperature is $5^\circ C$ when a system failure causes inside room temperature to drop from 20° to 17° in half an hour. After how many hours is the inside room temperature equal to $10^\circ C$? Give answer correct to 1 decimal place. 3

d) Let point $P(4p, 2p^2)$ be an arbitrary point on the parabola $x^2 = 8y$ with parameter p .

- i) Show that the equation of the tangent at P is $y = px - 2p^2$ 1
- ii) The tangent intersects the y -axis at C . The point Q divides CP , internally, in the ratio 1:3. Find the locus of all the points Q as parameter p varies. 3

Question 14 (15 marks) Use a SEPARATE writing booklet

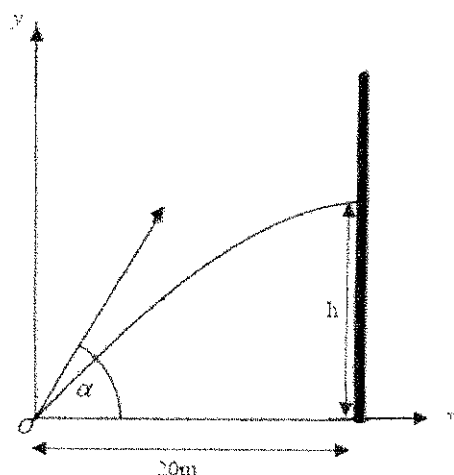
- a) Steven borrows \$50 000 to pay for a new car. He plans to repay the loan by making 60 equal monthly instalments. Interest is charged at the rate of 0.6% per month on the balance owing.
- i) Show that immediately after making two monthly instalments of \$M, the balance owing is given by $\$(50601.80 - 2.006M)$ 1
- ii) Calculate the value of each monthly instalment. 2
- b) Grain is poured at a constant rate of 0.5 cubic metres per second. It forms a conical pile, with the angle at the apex of the cone equal to 60° . The height of the pile is h metres, and the radius of the base is r metres.



- i) Show that $r = \frac{h}{\sqrt{3}}$ 1
- ii) Show that V , the volume of the pile, is given by $V = \frac{\pi h^3}{9}$. 1
- iii) Hence find the rate at which the height of the pile is increasing when the height of the pile is 3 metres. 2

Question 14 continued on next page

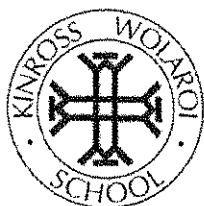
- c) A softball player hits the ball from ground level with a speed of 20 m/s and an angle of elevation α . It flies towards a high wall 20 m away on level ground. Taking the origin at the point where the ball is hit, the derived expressions for the horizontal and vertical components of x and y of displacement at the time t seconds, taking $g = 10 \text{ m/s}^2$, are $x = 20t \cos \alpha$ and $y = -5t^2 + 20t \sin \alpha$



- i) Hence find the equation of the path of the ball in flight in terms of x , y and α . 1
 - ii) Show that the height h at which the ball hits the wall is given by 1

$$h = 20 \tan \alpha - 5(1 + \tan^2 \alpha)$$
 - iii) Using part ii) above, show that the maximum value of h occurs when $\tan \alpha = 2$ and find this maximum height. 2
- d) Using the expansion of $(1+x)^n$ prove:
- i)
$$10^n = \binom{n}{0} + 3^2 \binom{n}{1} + 3^4 \binom{n}{2} + \dots + 3^{2n} \binom{n}{n}$$
 1
 - ii) Hence show that $1 + 3^4 \binom{n}{2} + 3^8 \binom{n}{4} + \dots + 3^{2n} \binom{n}{n} = 2^{n-1} (5^n + 4^n)$, 3
 where n is an even integer.

End of Examination



2015

Number: _____

Year 12 Extension 1 Mathematics

Trial Examination: Multiple Choice Answer Sheet

For multiple choice questions, choose the best answer A, B, C or D and fill in the correct circle.

1. ☐ A ☐ B ☐ C ☐ D
2. ☐ A ☐ B ☐ C ☐ D
3. ☐ A ☐ B ☐ C ☐ D
4. ☐ A ☐ B ☐ C ☐ D
5. ☐ A ☐ B ☐ C ☐ D
6. ☐ A ☐ B ☐ C ☐ D
7. ☐ A ☐ B ☐ C ☐ D
8. ☐ A ☐ B ☐ C ☐ D
9. ☐ A ☐ B ☐ C ☐ D
10. ☐ A ☐ B ☐ C ☐ D

Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Extension 1 2015 Trial

Section 1 Multiple choice

(1) $\int x \sqrt{1-x^2} dx$

$u = 1-x^2$

$du = -2x dx$

$= -\frac{1}{2} \int u^{1/2} du$

$-\frac{1}{2} du = x dx$

$= -\frac{1}{2} \times \frac{2}{3} u^{3/2} + C$

$= -\frac{1}{3} u^{3/2} + C$

$= -\frac{1}{3} (1-x^2)^{3/2} + C$

(C)

(2) $\int \sin^2 6x dx$

$= \int \frac{1}{2} - \frac{1}{2} \cos 12x dx$

$= \frac{x}{2} - \frac{1}{24} \sin 12x + C$

(C)

(3) $V = \sqrt{8x-x^2}$

$V^2 = 8x-x^2$

$\frac{1}{2} V^2 = 4x - \frac{1}{2} x^2$

$\frac{d}{dx} \left(\frac{1}{2} V^2 \right) = 4 - x$

$\ddot{x} = 4 - x$

when $x=3$

$x=1$

(A)

(4) ${}^5C_2 \times {}^{10}C_6$

There are 5 prefects to choose 2 from 5C_2

leaving 10 students to choose the other 6

${}^{10}C_6$

(A)

(5) $x = 3 \cos 2t + 4 \sin 2t$

greatest speed occurs at centre of motion

$x = 5 \cos (2t - \tan^{-1}(\frac{4}{3}))$

$\dot{x} = -10 \sin (2t - \tan^{-1}(\frac{4}{3}))$

max speed is 10 (B)

(6) $y = e^{x+2}$

$x = e^{y+2}$

$\ln x = y+2$

$\ln x - 2 = y$

(D)

(7) $y = \cos^{-1} \frac{3x}{2}$

Domain $-1 \leq \frac{3x}{2} \leq 1$

$-\frac{2}{3} \leq x \leq \frac{2}{3}$

Range $0 \leq y \leq \pi$

(A)

(8) $\int_{2/\sqrt{3}}^{2\sqrt{3}} \frac{dx}{x^2+4}$

$\left[\frac{1}{2} \tan^{-1} \frac{x}{2} \right]_{2/\sqrt{3}}^{2\sqrt{3}}$

$= \frac{1}{2} \left[\tan^{-1} \sqrt{3} - \tan^{-1} \frac{1}{\sqrt{3}} \right]$

$= \frac{1}{2} \left[\frac{\pi}{3} - \frac{\pi}{6} \right]$

$= \frac{1}{2} \cdot \frac{\pi}{6}$

$= \frac{\pi}{12}$

(A)

(9) $(x^2 - 2/x)^9$

${}^9C_k (x^2)^{9-k} \left(\frac{-2}{x} \right)^k$

${}^9C_k x^{18-2k} (-2)^k x^{-k}$

${}^9C_k x^{18-3k} (-2)^k$

$\therefore 18-3k=0$

$k=6$

${}^9C_6 (-2)^6$

(B)

⑩ $f(x) = \sin x - \frac{2x}{3}$

$f'(x) = \cos x - \frac{2}{3}$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 1.5 - \frac{\sin 1.5 - \frac{2}{3} \times 1.5}{\cos 1.5 - \frac{2}{3}}$$

$$= 1.49579 \dots$$

$$\approx 1.496$$

(B)

Section 2

Question 11

a) $\frac{4}{5-x} > 1$

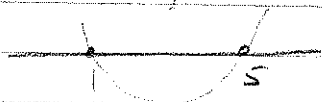
$4(5-x) > (5-x)^2$

$0 > (5-x)^2 - 4(5-x)$

$0 > (5-x)(5-x-4)$

$0 > (5-x)(1-x)$

C.V. $x = 5, 1$ ✓



$1 \leq x < 5$

b) $A(1, 4) \xrightarrow{-1:3} B(5, 2)$

$x = \frac{3(1) + (5) - 1}{2} = -1$

$y = \frac{3(4) + (-1) \cdot 2}{2} = 5$

$\therefore (-1, 5)$

c) i) DELETED

$$\frac{7!}{3! \cdot 2!} = \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3}{3 \cdot 2}$$

$$= 420$$

ii) Consider the D's together

$\text{DD} \text{ --- }$

$\frac{6!}{3!} = 120$

$\therefore$ the number of words with the D's separated is

$420 - 120 = 300$

d) $\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$ $m_1 = 2$ $m_2 = -1$

$= \left| \frac{2 + 1}{1 - 2} \right|$

$= \left| -3 \right|$

$= 3$

$\theta = 71^\circ 34'$

e) $P(-2) = 0$

$(-2)^3 - 2(-2)^2 + (-2)a + 4 = 0$

$-8 - 8 - 2a + 4 = 0$

$a = -6$

f) $\alpha + \beta + \gamma = \frac{2}{3}$

$\alpha\beta + \beta\gamma + \alpha\gamma = 1$

$\alpha\beta\gamma = \frac{4}{3}$

$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\alpha\gamma} = \frac{\gamma + \alpha + \beta}{\alpha\beta\gamma}$

$= \frac{\frac{2}{3}}{\frac{4}{3}}$

$= \frac{1}{2}$

Question 12:

a) i) $\frac{d}{dx} \ln(\cos 2x)$
 $= \frac{1}{\cos 2x} \cdot -2 \sin 2x$

$= -2 \tan 2x$ ✓

ii) $\therefore \int -2 \tan 2x \, dx = \ln(\cos 2x) + c$

$-2 \int_0^{\pi/6} \tan 2x \, dx = \left[\ln(\cos 2x) \right]_0^{\pi/6}$

$= -\frac{1}{2} (\ln(\cos \frac{\pi}{3}) - \ln(\cos 0))$

$= -\frac{1}{2} (\ln(\frac{1}{2}) - \ln(1))$

$= -\frac{1}{2} \ln(\frac{1}{2}) = \frac{1}{2} \ln(\frac{1}{2}) = \frac{1}{2} \ln 2$

b) $T_{n+1} - T_n = 7$ $T_1 = 3$

$T_2 - T_1 = 7$

$T_2 = 10$

$\therefore AP \ a = 3 \ d = 7$ ✓

$S_n = \frac{n}{2} (a + l)$ $T_{100} = 3 + 99 \times 7$
 $= 696$

$S_{100} = 50 (3 + 696)$
 $= 34950$ ✓

c) i) $\cos x - \sqrt{3} \sin x$

$r = \sqrt{3+1}$ $\tan \alpha = \sqrt{3}$

$= 2$

$\alpha = \pi/3$

$\therefore \cos x - \sqrt{3} \sin x = 2 \cos(x + \frac{\pi}{3})$ ✓

ii) $2 \cos(x + \frac{\pi}{3}) = 1$

$\cos(x + \frac{\pi}{3}) = \frac{1}{2}$

$x + \pi/3 = 2\pi \pm \cos^{-1}(\frac{1}{2})$

$= 2\pi \pm \frac{\pi}{3}$ ✓

$= 2\pi \pm \frac{\pi}{3}$ ✓

d) show true for $n=1$

LHS = $\frac{1}{5}$ RHS = $\frac{1}{4+1}$
 $= \frac{1}{5}$ ✓

$\therefore$ true for $n=1$

assume true for $n=k$

$S_k = \frac{k}{4k+1}$

Prove true for $n=k+1$

$S_k + T_{k+1} = \frac{k+1}{4(k+1)+1}$

$= \frac{k+1}{4k+5}$

LHS = $\frac{k}{4k+1} + \frac{1}{(4(k+1)+1)}$ ✓

$= \frac{k}{4k+1} + \frac{1}{(4k+5)}$

$= \frac{-k(4k+5) + 1}{(4k+1)(4k+5)}$

$= \frac{4k^2 + 5k + 1}{(4k+1)(4k+5)}$

$= \frac{(4k+1)(k+1)}{(4k+5)(4k+1)}$ ✓

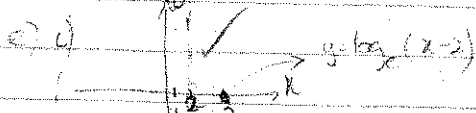
$= \frac{k+1}{4k+5}$

$\therefore$ True for $n=1$, whenever

true for $n=k$ it is true for

$n=k+1$ $\therefore$ true for all integers

$n \geq 1$



iii) $V = \pi \int_0^h x^2 \, dy$ $e^y = x-2$

$e^y - 2 = x$

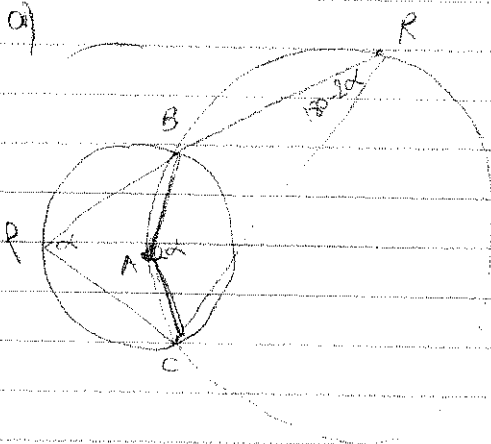
$V = \pi \int_0^h (e^{2y} - 4e^y + 4) \, dy$ ✓

$= \pi \left[\frac{e^{2y}}{2} - 4e^y + 4y \right]_0^h$ ✓

$= \pi \left[\frac{e^{2h}}{2} - 4e^h + 4h - \left(\frac{1}{2} - 4 + 0 \right) \right]$ ✓

Question B:

a)



$$\tan 15^\circ = \frac{200}{a}$$

$$\tan 10^\circ = \frac{200}{b}$$

$$a = \frac{200}{\tan 15^\circ}$$

$$b = \frac{200}{\tan 10^\circ}$$

using cosine rule

$$x^2 = \frac{200^2}{\tan^2 15^\circ} + \frac{200^2}{\tan^2 10^\circ} - 2 \times \frac{200}{\tan 15^\circ} \times \frac{200}{\tan 10^\circ} \cos 10^\circ$$

$$= 21376.93834$$

$$x = 146.2 \text{ m}$$

$$\text{Let } \angle PBC = x$$

$\therefore \angle BAC = 2x$ (angle at centre is twice the angle on the circumference)

$\angle BRC = 180 - 2x$ (ACBR is cyclic $\therefore$ opposite angles supplementary)

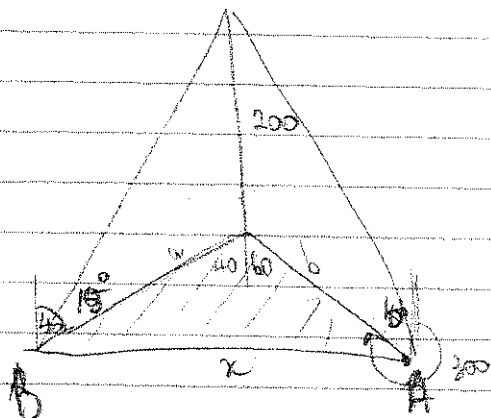
In $\triangle PRC$

$$\angle PCR = 180 - x - (180 - 2x) = x$$

$\therefore \triangle PRC$ is isosceles (base angles equal)

$\therefore PR = PC$ (equal sides of isosceles $\triangle$)

b)



c) i) $T = A + Ce^{kt}$

$$\frac{dT}{dt} = kCe^{kt}$$

$$= k(Ce^{kt} - A + A)$$

$$= k(T - A)$$

ii) $T = A + Ce^{kt}$

$$T = 5 + Ce^{kt}$$

when $t = 0, T = 20$

$$\therefore C = 15$$

$$T = 5 + 15e^{kt}$$

when $T = 17, t = 0.5$

$$17 = 5 + 15e^{0.5k}$$

$$\log_e \left(\frac{12}{15} \right) = 0.5k$$

$$-0.446287 = k$$

let $T = 10$

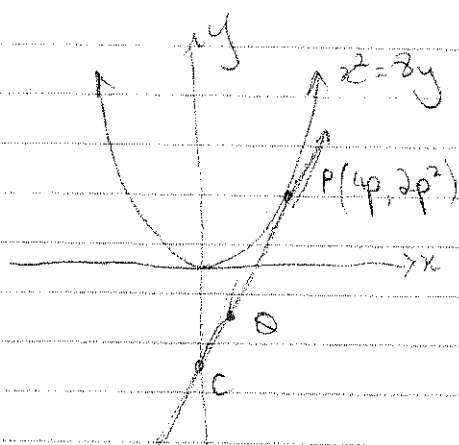
$$10 = 5 + 15e^{kt}$$

$$\log_e \left(\frac{5}{15} \right) = kt$$

$$\therefore t = 2.46$$

$$\approx 2.5 \text{ hours}$$

d)



$$i) y = \frac{x^2}{8}$$

$$y' = \frac{x}{4}$$

$$m = \frac{4p}{4} = p$$

$$y - 2p^2 = p(x - 4p)$$

$$y - 2p^2 = px - 4p^2$$

$$y = px - 2p^2$$

as required.

$$ii) C(0, -2p^2) \quad (4p, 2p^2)$$

$$x = \frac{4p}{4} = p$$

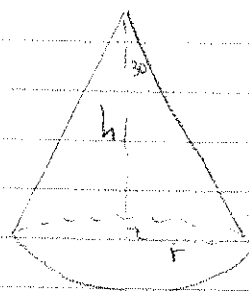
$$y = \frac{-6p^2 + 2p^2}{4} = \frac{-4p^2}{4} = -p^2$$

$$\therefore \text{locus is } y = x^2$$

Question 14:

a) is on next pg.

b)



$$i) \tan 30 = \frac{r}{h}$$

$$h \times \frac{1}{\sqrt{3}} = r$$

$$h = r\sqrt{3}$$

$$ii) V = \frac{1}{3} \pi r^2 h$$

$$= \frac{1}{3} \pi \frac{h^2}{3} \times h$$

$$= \frac{1}{9} \pi h^3$$

$$iii) \frac{dV}{dt} = 0.5$$

$$V = \frac{1}{9} \pi h^3$$

$$\frac{dV}{dh} = \frac{1}{3} \pi h^2$$

$$\text{Find } \frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$

$$= \frac{3}{\pi h^2} \times 0.5$$

$$= \frac{3}{\pi \times 93} \times 0.5$$

$$= \frac{1}{6\pi}$$

$$c) i) x = 20t \cos \alpha \quad y = -5t^2 + 20t \sin \alpha$$

$$y = -5 \left(\frac{x}{20 \cos \alpha} \right)^2 + 20 \left(\frac{x}{20 \cos \alpha} \right) \sin \alpha$$

$$= \frac{-5x^2}{400 \cos^2 \alpha} + x \tan \alpha$$

$$= -\frac{x^2}{80} \sec^2 \alpha + x \tan \alpha$$

ii) when $x = 20$ $y = h$

$$h = -\frac{20^2}{80} (1 + \tan^2 \alpha) + 20 \tan \alpha$$

$$= -\frac{400}{80} (1 + \tan^2 \alpha) + 20 \tan \alpha$$

$$= 20 \tan \alpha - 5(1 + \tan^2 \alpha)$$

$$iii) h = -5 \tan^2 \alpha + 20 \tan \alpha - 5$$

maximum value of h occurs

at the turning point

$$\tan \alpha = -b/2a$$

$$= \frac{-20}{2(-5)}$$

$$= 2$$

maximum height is

$$-5(2)^2 + 20(2) - 5$$

$$= 15 \text{ metres}$$

d)

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + {}^nC_3 x^3 + \dots + {}^nC_n x^n$$

let $x=9$

$$10^n = {}^nC_0 + {}^nC_1 9 + {}^nC_2 9^2 + {}^nC_3 9^3 + \dots + {}^nC_n 9^n$$

$$= {}^nC_0 + {}^nC_1 3^2 + {}^nC_2 3^4 + \dots + {}^nC_n 3^{2n}$$

$$= \binom{n}{0} + 3 \binom{n}{1} + 3^2 \binom{n}{2} + \dots + 3^n \binom{n}{n} \quad \text{--- (1)}$$

ii)

$$\text{let } x = -9$$

$$8^n = {}^nC_0 - {}^nC_1 3 + {}^nC_2 3^2 - {}^nC_3 3^3 + \dots + {}^nC_n (-9)^n \quad \text{--- (2)}$$

Add (1) + (2)

$$10^n + 8^n = 2 {}^nC_0 + 2 {}^nC_2 3^2 + \dots + 2 {}^nC_n 3^n$$

$$2^n 5^n + 2^n 4^n = 2 \left(\binom{n}{0} + 3^2 \binom{n}{2} + \dots + 3^n \binom{n}{n} \right)$$

$$2^n (5^n + 4^n) = 2 \left(\binom{n}{0} + 3^2 \binom{n}{2} + \dots + 3^n \binom{n}{n} \right)$$

$$2^{n-1} (5^n + 4^n) = \binom{n}{0} + 3^2 \binom{n}{2} + 3^3 \binom{n}{3} + \dots + 3^{n-1} \binom{n}{n}$$

$$a) A_1 = 50000 \times 1.006 - M$$

$$A_2 = (50000 \times 1.006 - M) \times 1.006 - M$$

$$= 50000 \times 1.006^2 - M(1.006) - M$$

$$= 50601.80 - M \times 2.006$$

$$A_3 = 50000 \times 1.006^3 - M(1.006^2 + 1.006 + 1)$$

$$A_{60} = 50000 \times 1.006^{60} - M(1 + 1.006 + \dots + 1.006^{59})$$

but $A_n = 0$ when paid off. $n=60$

$$0 = 50000 \times 1.006^{60} - M(1 + 1.006 + \dots + 1.006^{59})$$

$$= 50000 \times 1.006^{60} - M \left(1 + \frac{1.006^{60} - 1}{1.006 - 1} \right)$$

$$M = \frac{50000 \times 1.006^{60} \times (1.006 - 1)}{1.006^{60} - 1}$$

$$= \$994.78$$